

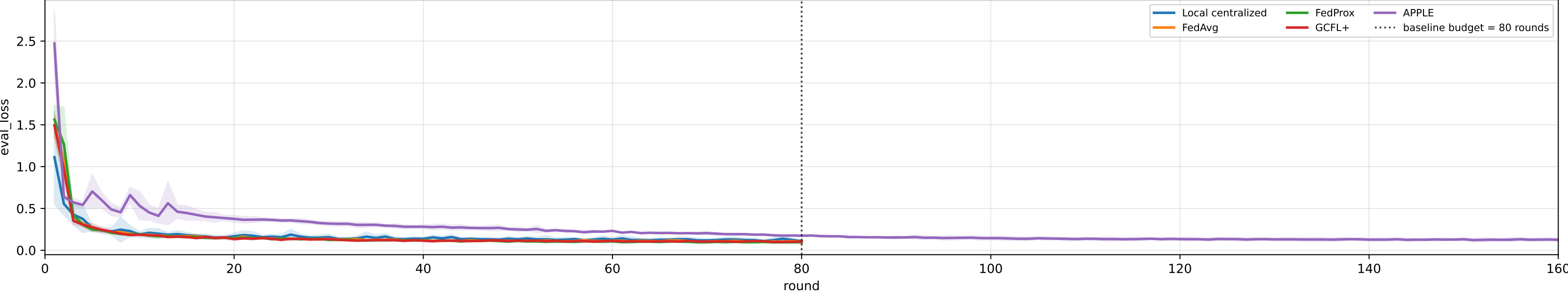
Controlled task-specialization level: Low | aggregated / pooled performance over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
family=specialized_cycle4 | n_clients=1 | cycle2=visible 5.1%±0.0% / true 6.4% | cycle3=visible 5.0%±0.0% / true 6.3% | cycle4=visible 5.3%±0.0% / true 6.6% |
family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |
family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.94 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

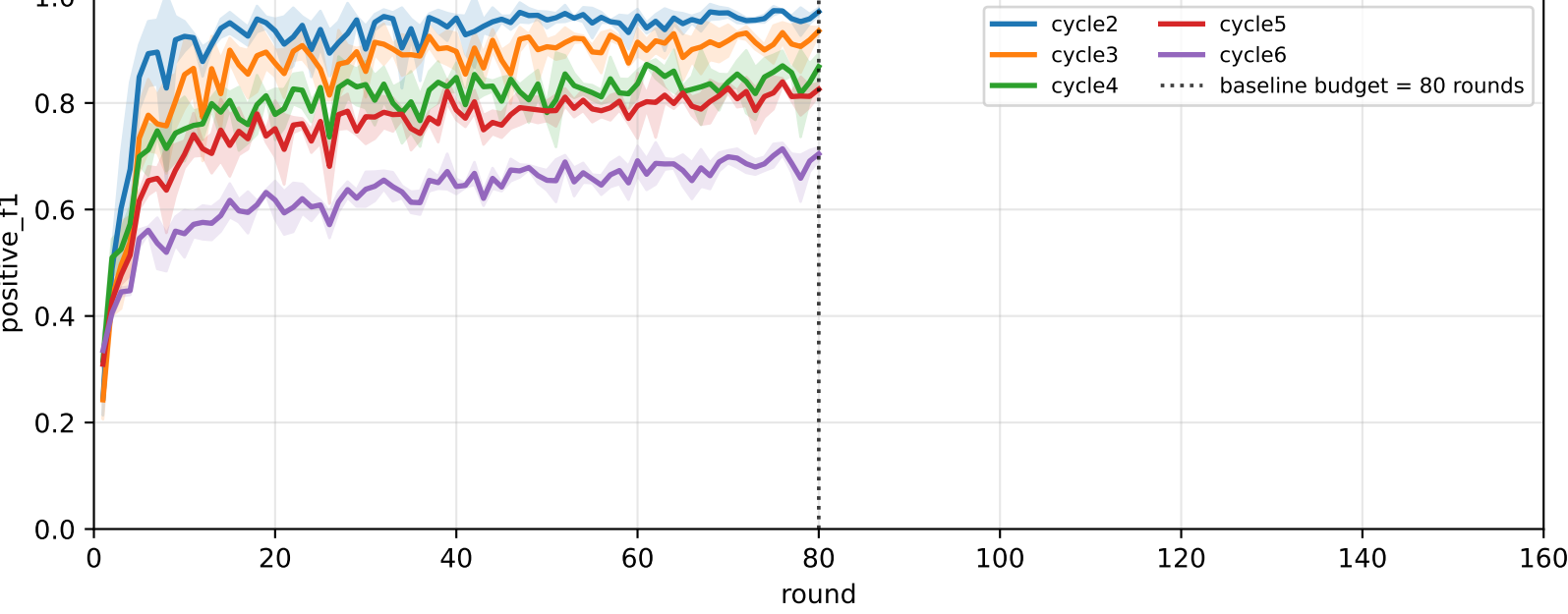
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	83.4% ± 0.3%	85.7% ± 0.3%	96.8% ± 0.6%	94.9% ± 0.4%	86.8% ± 0.2%	78.7% ± 0.7%	70.4% ± 0.6%
FedAvg	81.6% ± 0.3%	84.0% ± 0.3%	96.2% ± 0.8%	95.4% ± 0.7%	82.3% ± 2.8%	78.0% ± 0.2%	68.4% ± 2.8%
FedProx	83.5% ± 0.5%	85.3% ± 0.5%	95.4% ± 0.9%	95.1% ± 1.1%	85.3% ± 1.1%	77.9% ± 0.5%	73.0% ± 1.2%
GCFL+	82.0% ± 0.2%	84.0% ± 0.1%	95.3% ± 0.8%	93.7% ± 1.0%	83.5% ± 0.6%	77.3% ± 1.3%	70.3% ± 0.5%
APPLE	80.5% ± 2.0%	82.4% ± 1.9%	95.1% ± 1.2%	90.8% ± 2.4%	80.4% ± 2.8%	75.1% ± 1.2%	70.5% ± 2.5%

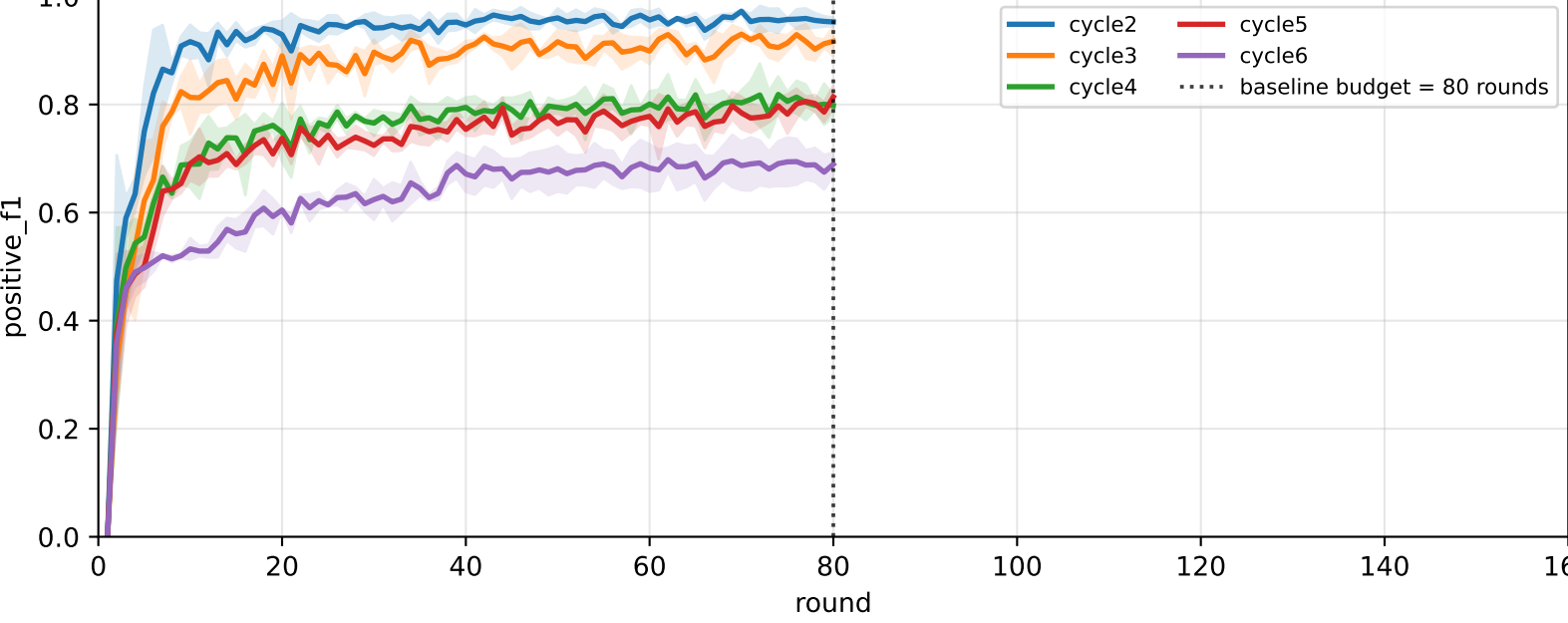
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



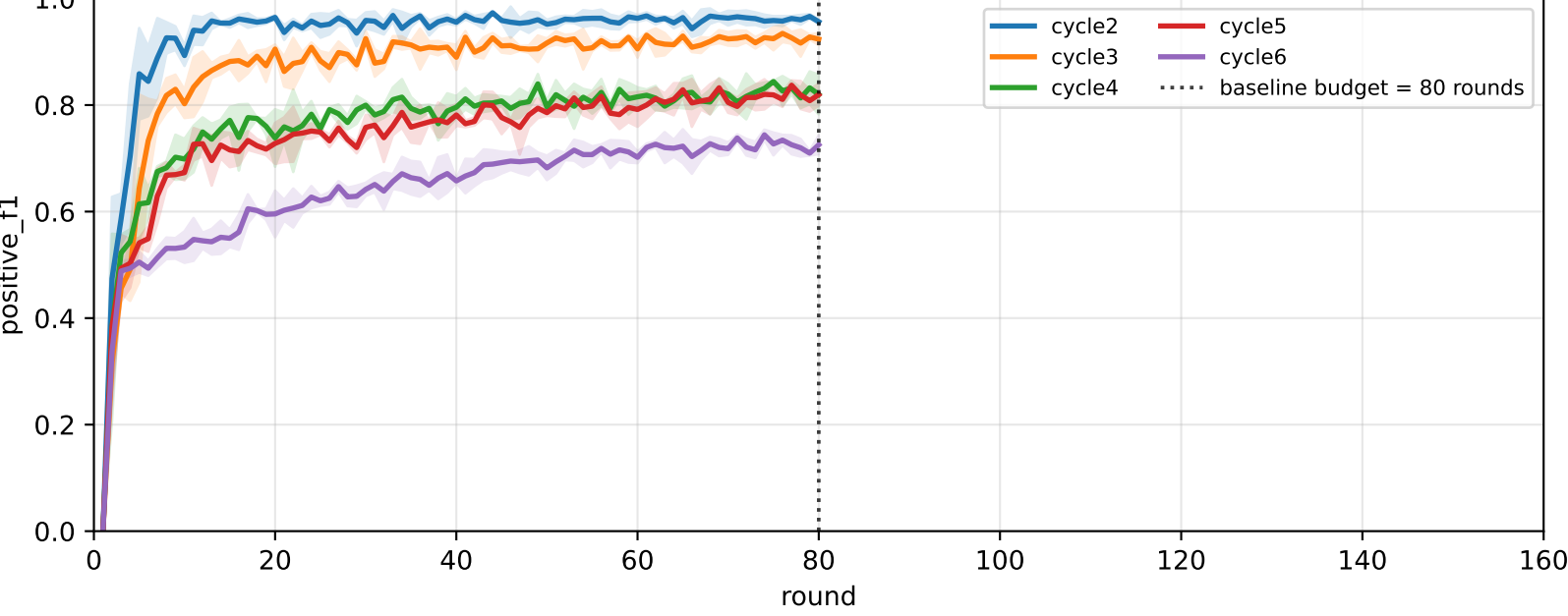
Pooled local centralized positive F1



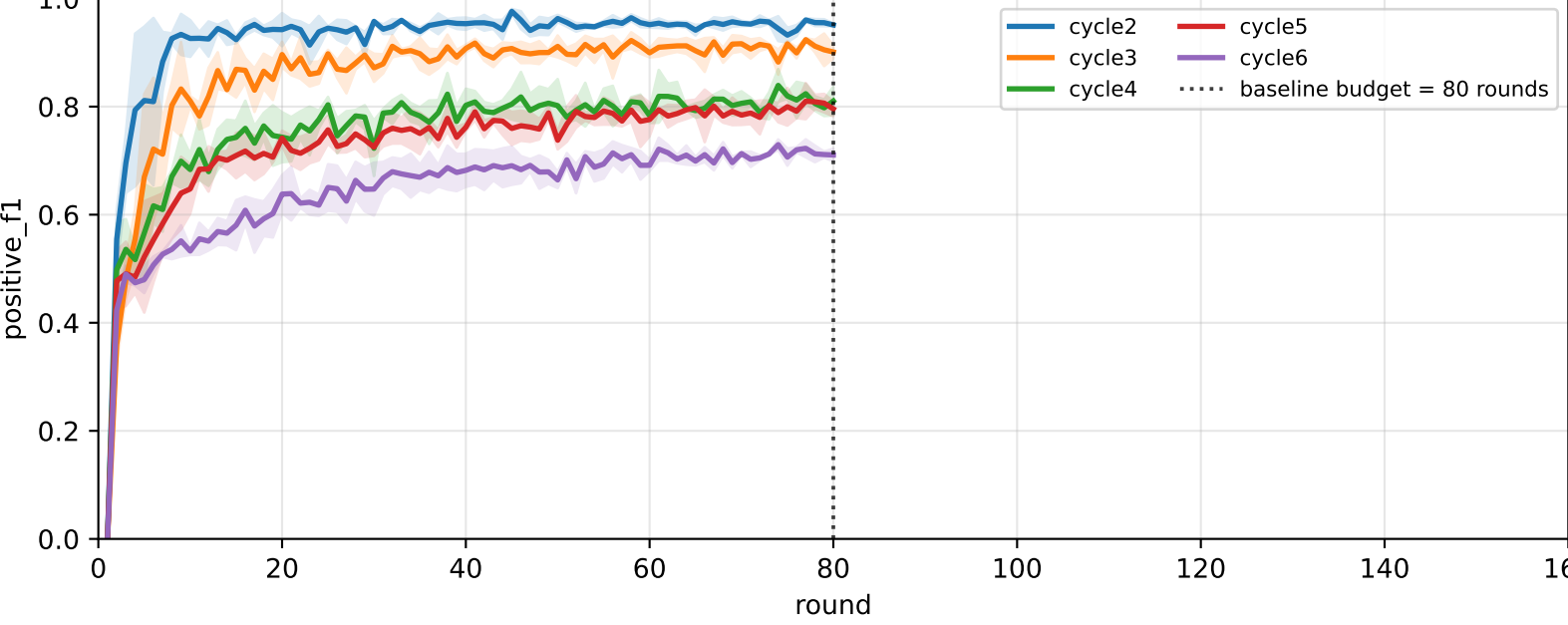
Pooled FedAvg positive F1



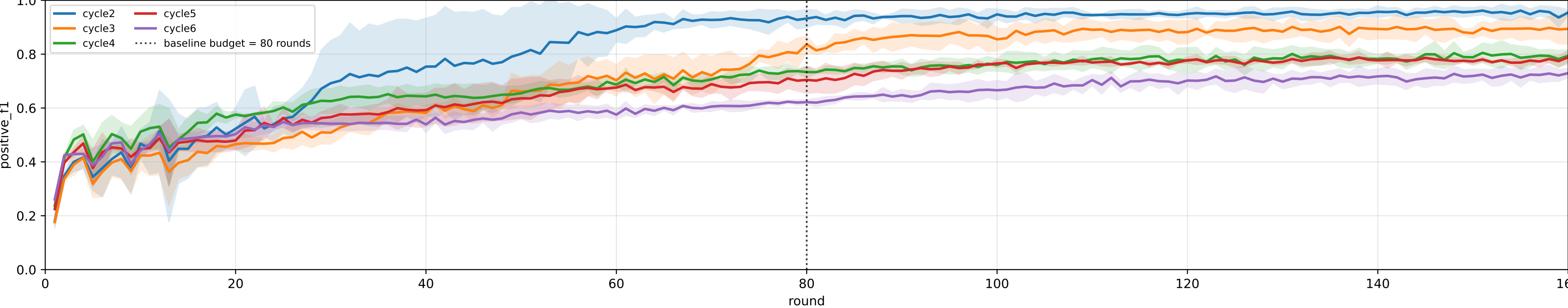
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



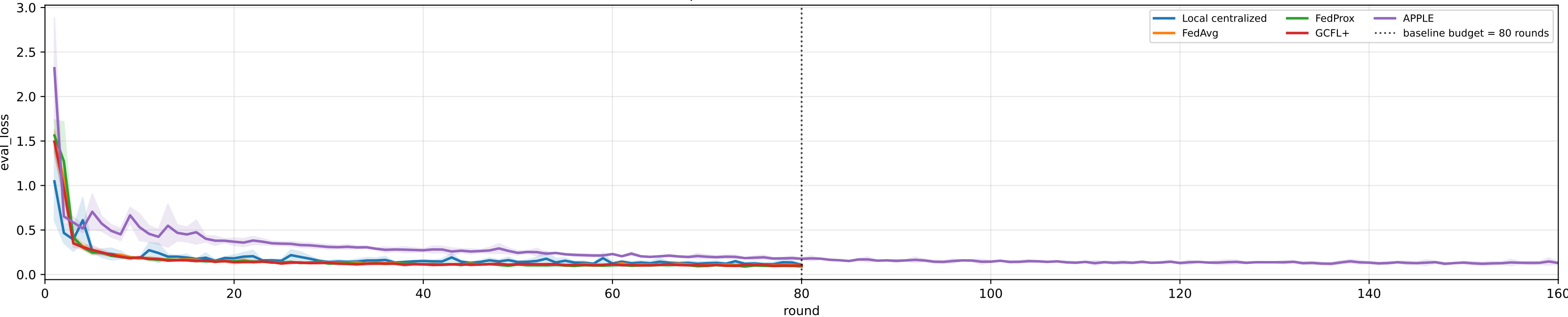
Controlled task-specialization level: Low | family aggregated / pooled performance |
family=specialized_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

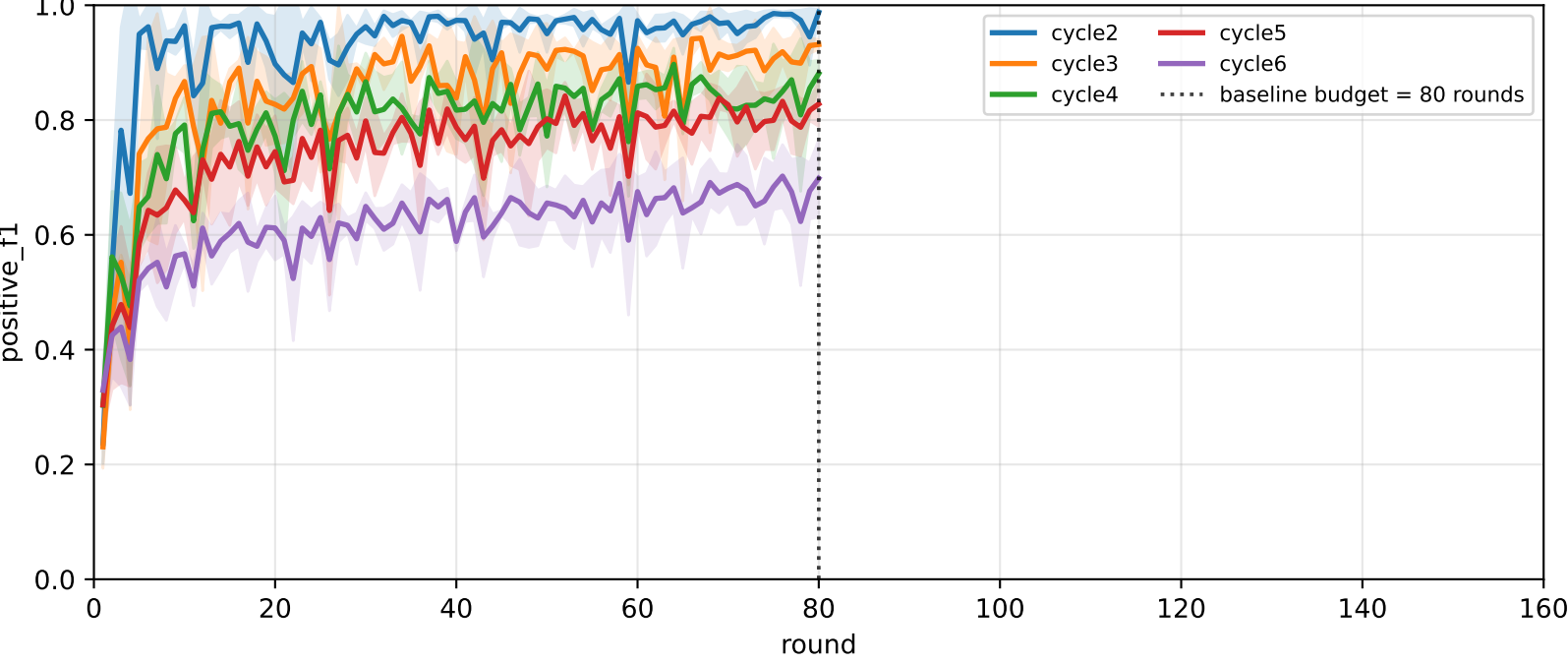
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.5% ± 1.5%	84.0% ± 1.5%	96.6% ± 1.3%	93.0% ± 1.5%	85.5% ± 2.0%	78.8% ± 2.1%	66.4% ± 2.0%
FedAvg	81.6% ± 0.2%	84.2% ± 0.5%	97.1% ± 1.0%	95.7% ± 0.5%	81.6% ± 3.0%	78.2% ± 0.7%	68.3% ± 2.6%
FedProx	83.8% ± 0.9%	85.6% ± 0.9%	96.3% ± 1.2%	95.6% ± 1.5%	85.1% ± 1.2%	77.8% ± 0.3%	73.3% ± 1.4%
GCFL+	82.4% ± 0.2%	84.5% ± 0.1%	95.9% ± 0.4%	93.7% ± 1.6%	84.5% ± 0.9%	77.5% ± 0.9%	70.8% ± 0.5%
APPLE	80.8% ± 1.7%	82.6% ± 1.7%	95.5% ± 2.1%	90.1% ± 2.9%	81.0% ± 1.8%	75.1% ± 1.0%	71.3% ± 1.8%

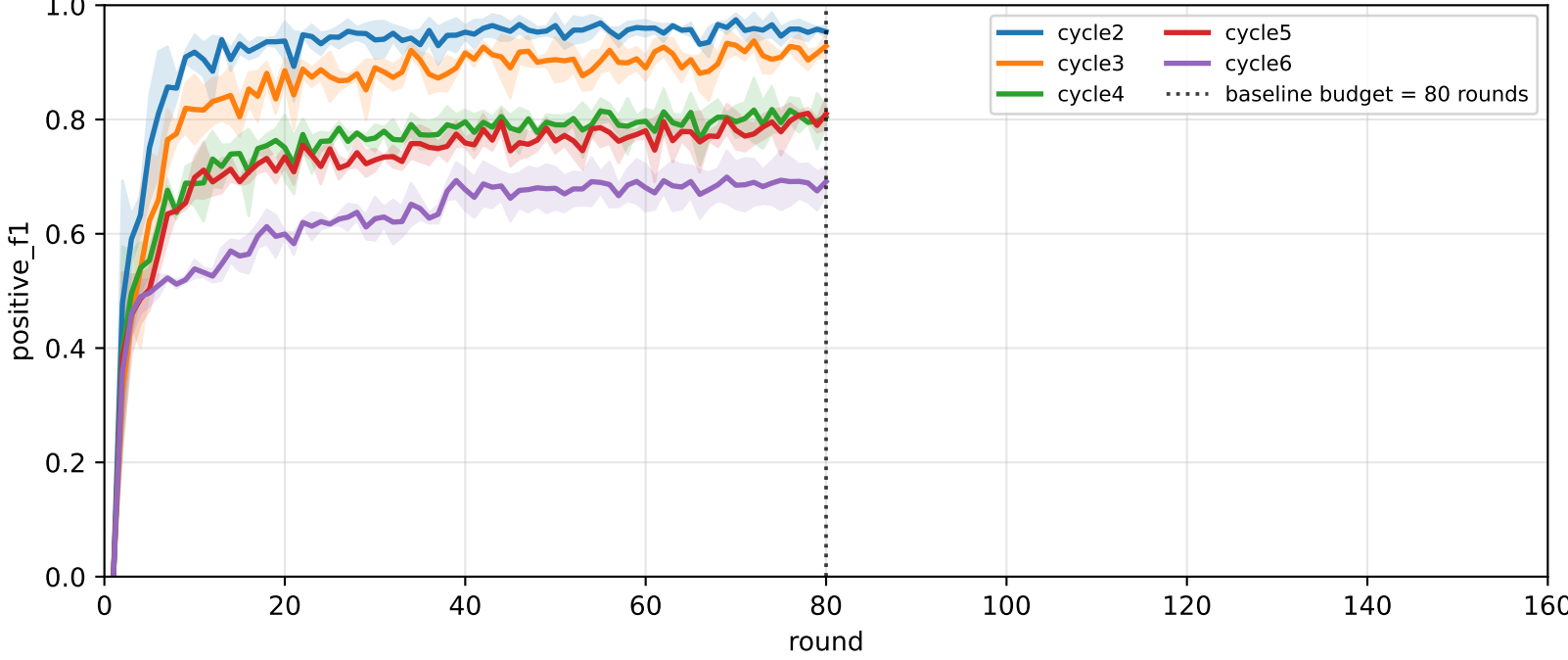
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



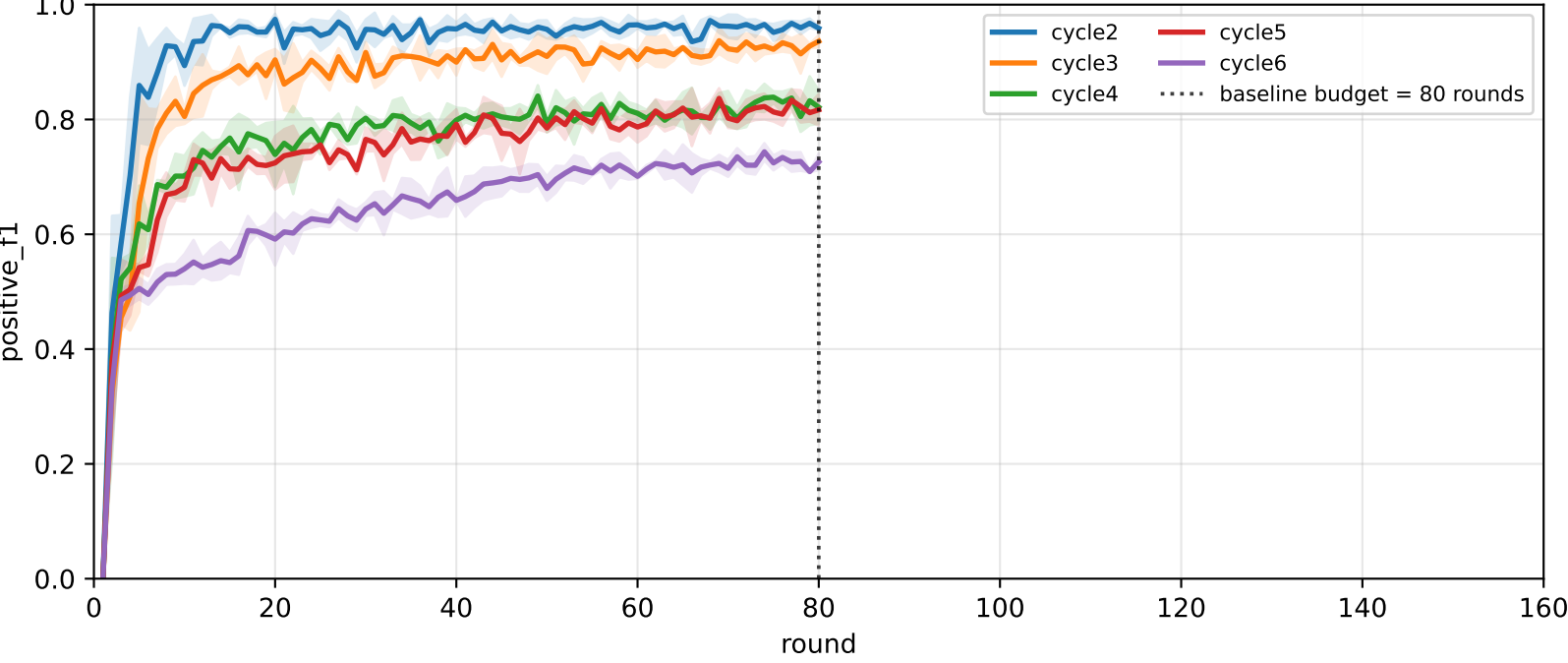
Pooled local centralized positive F1



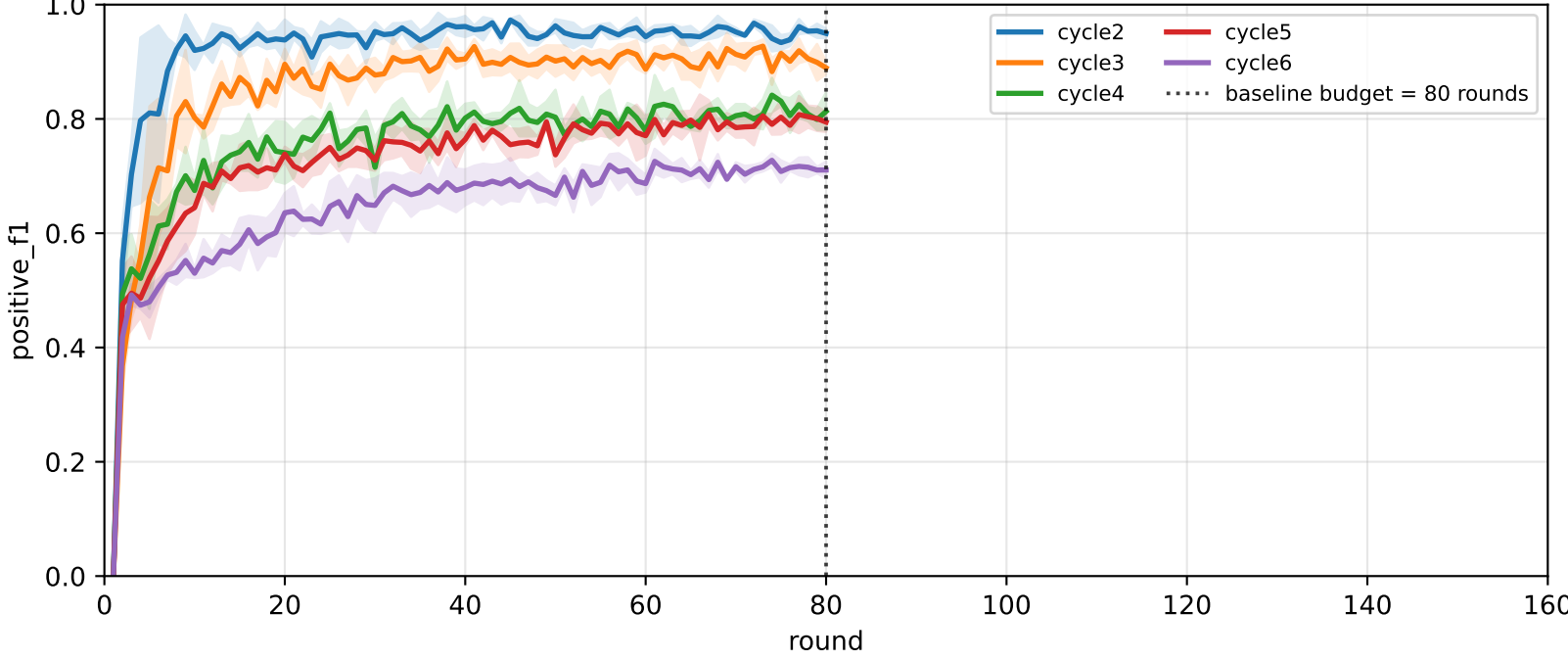
Pooled FedAvg positive F1



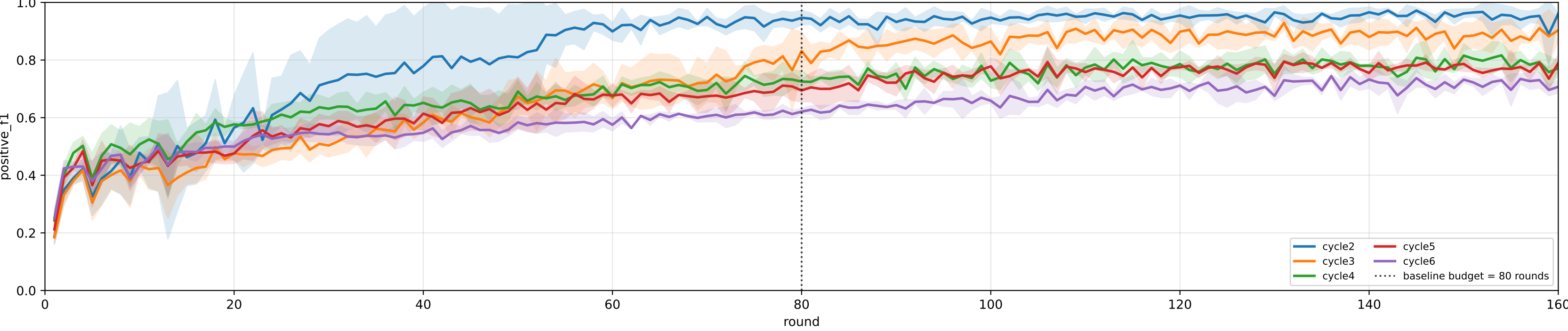
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



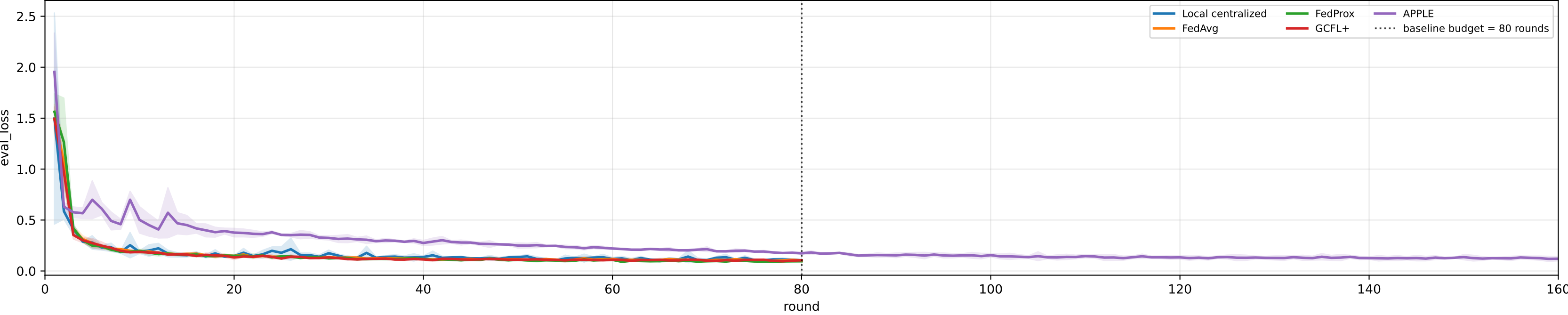
Controlled task-specialization level: Low | family aggregated / pooled performance |
family=specialized_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

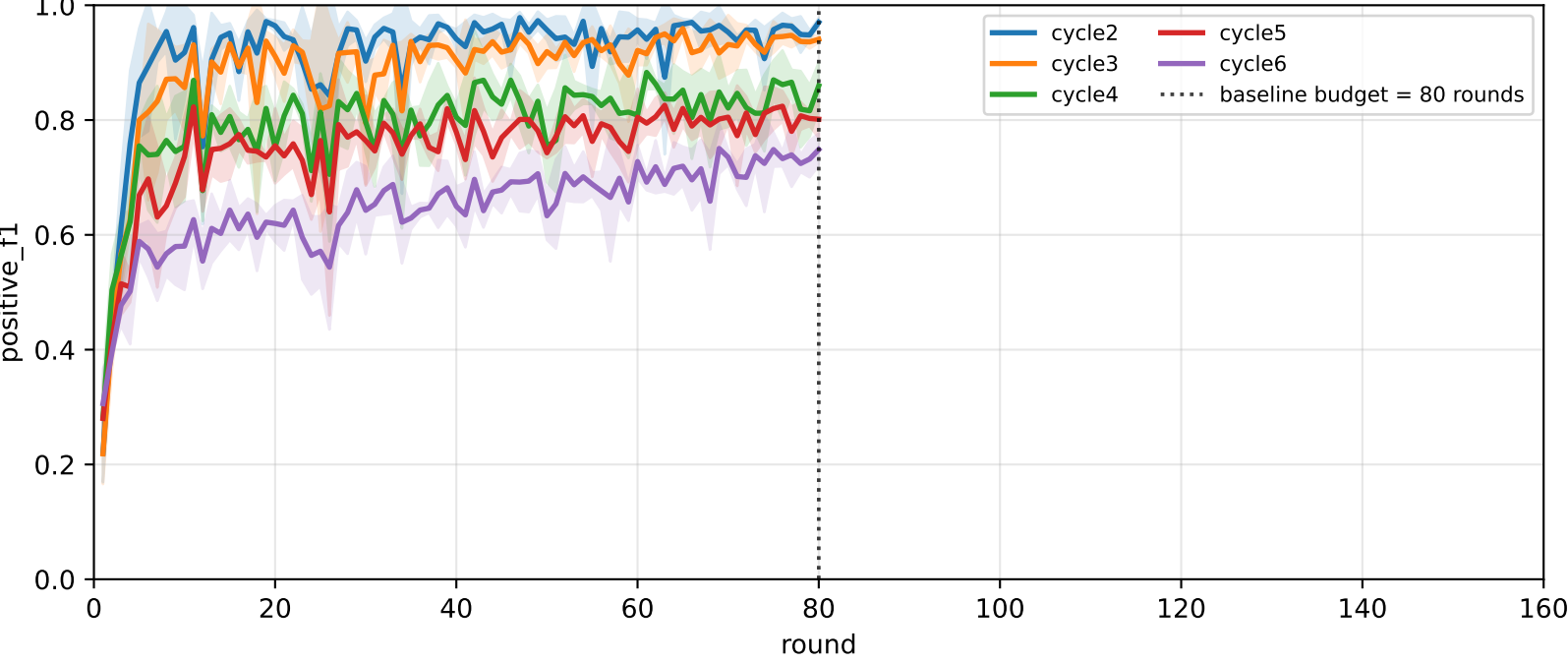
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.3% ± 1.4%	86.1% ± 1.1%	96.7% ± 0.5%	95.8% ± 1.1%	86.6% ± 2.6%	78.0% ± 2.0%	73.7% ± 3.9%
FedAvg	81.5% ± 0.2%	83.9% ± 0.5%	95.4% ± 0.8%	95.8% ± 1.0%	82.4% ± 3.4%	77.1% ± 0.4%	68.7% ± 2.8%
FedProx	83.5% ± 1.0%	85.3% ± 0.9%	95.2% ± 0.8%	95.3% ± 1.1%	85.1% ± 1.7%	77.8% ± 0.8%	73.1% ± 1.3%
GCFL+	82.3% ± 0.4%	84.3% ± 0.4%	95.2% ± 1.1%	94.3% ± 0.7%	83.9% ± 0.9%	77.6% ± 2.1%	70.4% ± 0.4%
APPLE	80.9% ± 2.4%	82.8% ± 2.1%	94.5% ± 0.7%	92.3% ± 2.8%	80.2% ± 2.8%	76.9% ± 1.0%	70.4% ± 3.8%

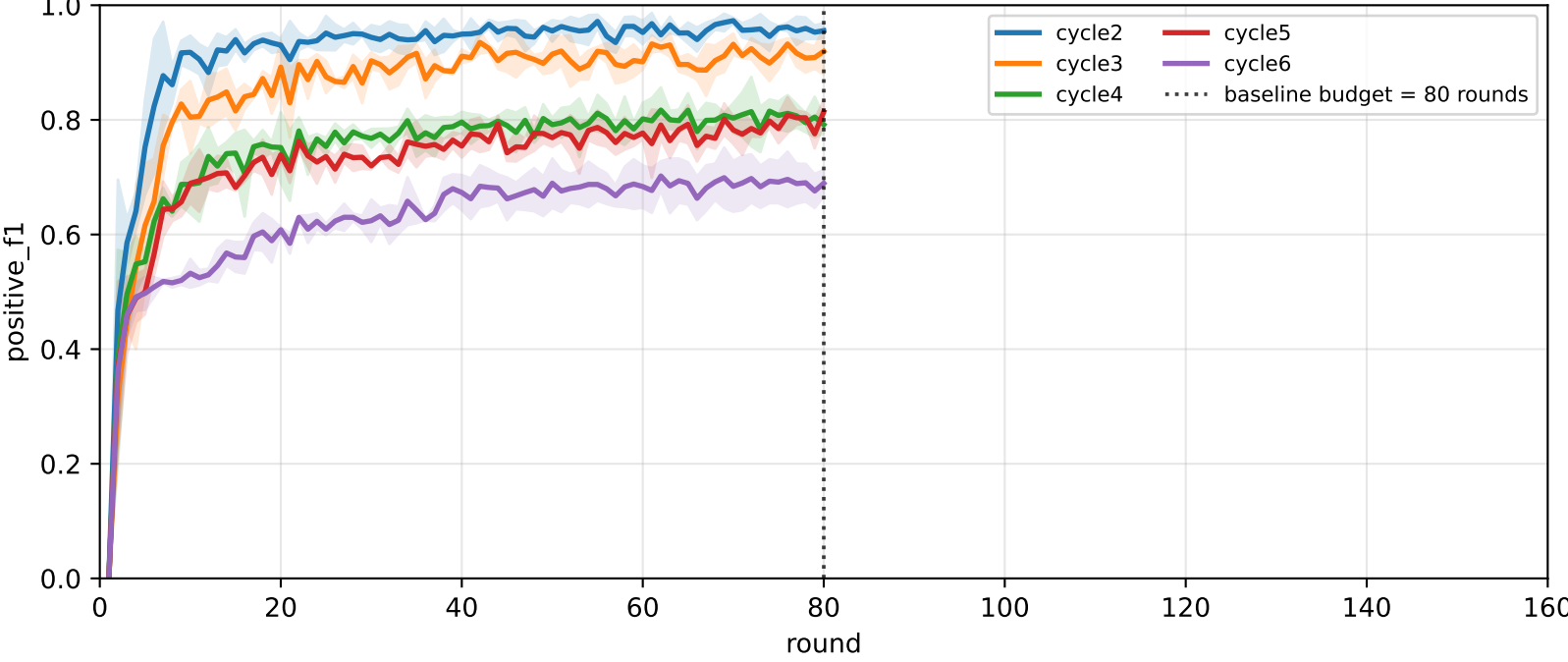
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



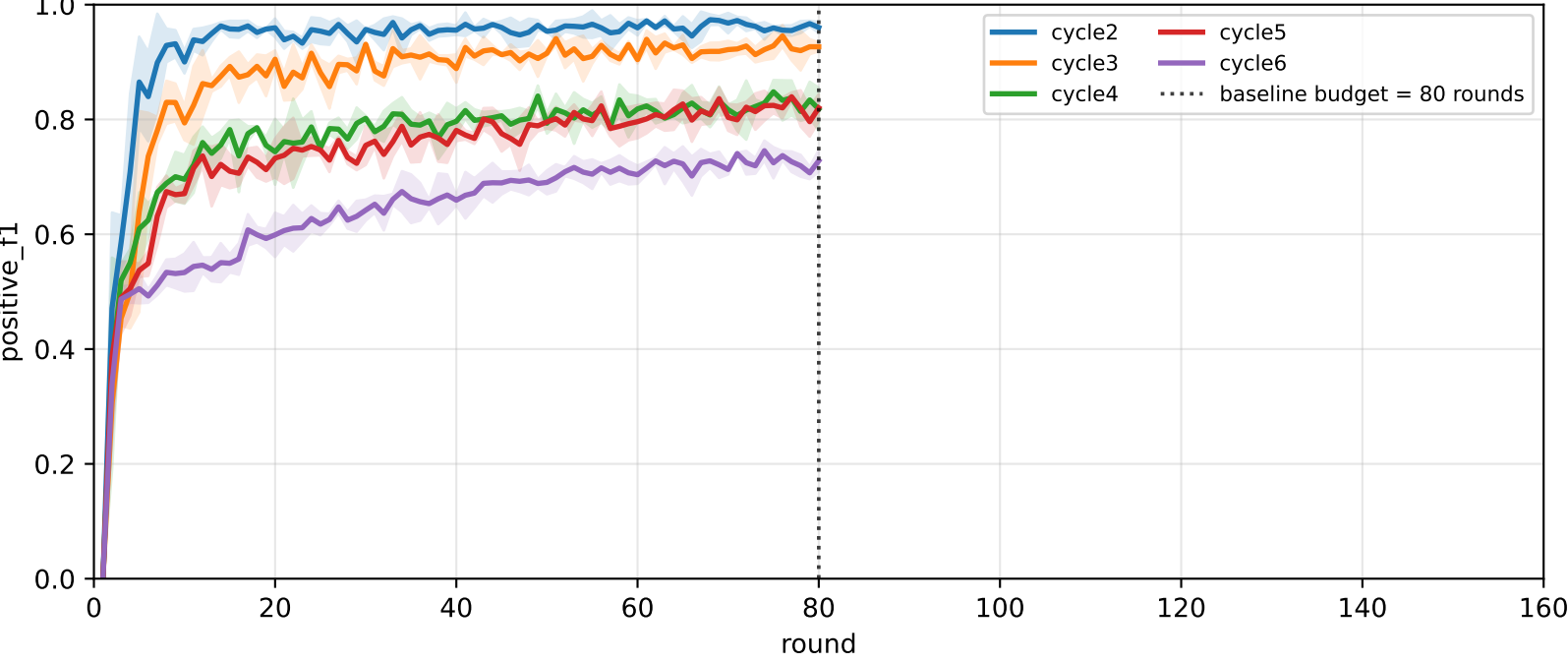
Pooled local centralized positive F1



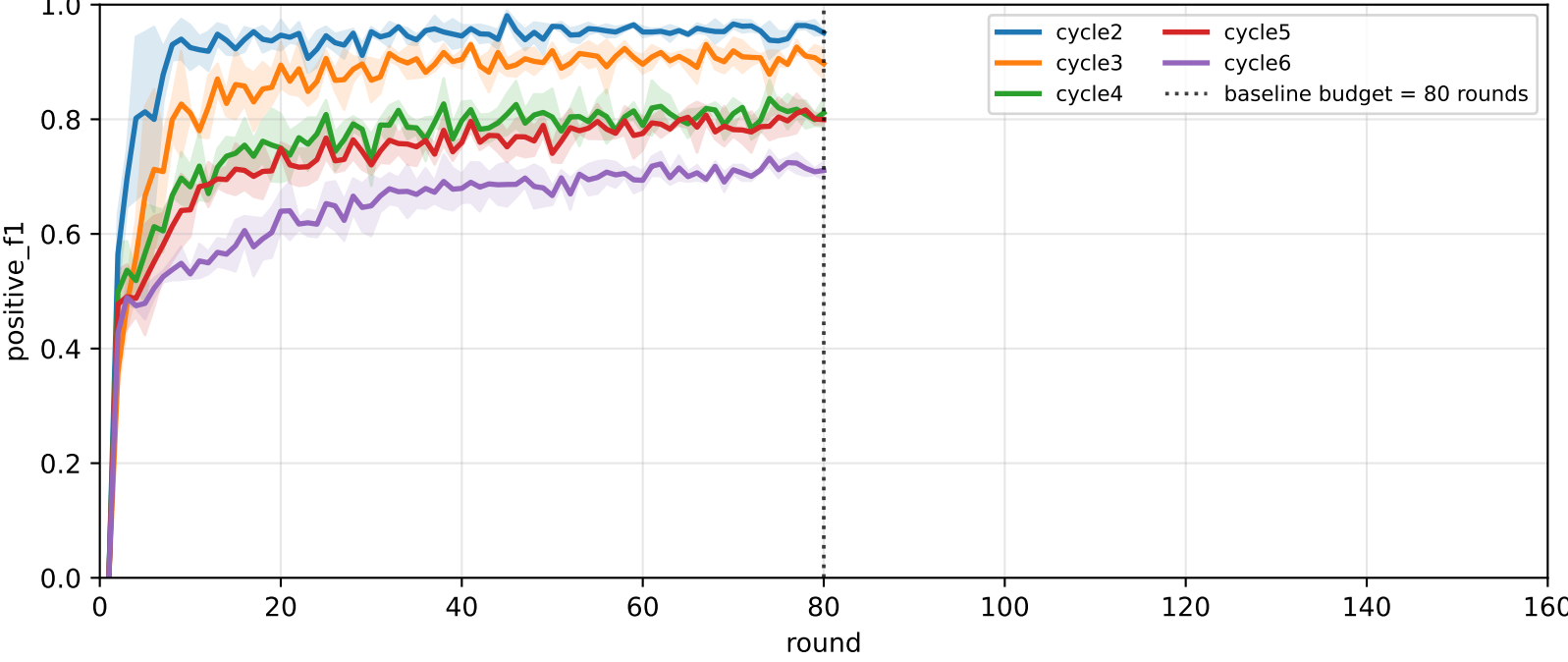
Pooled FedAvg positive F1



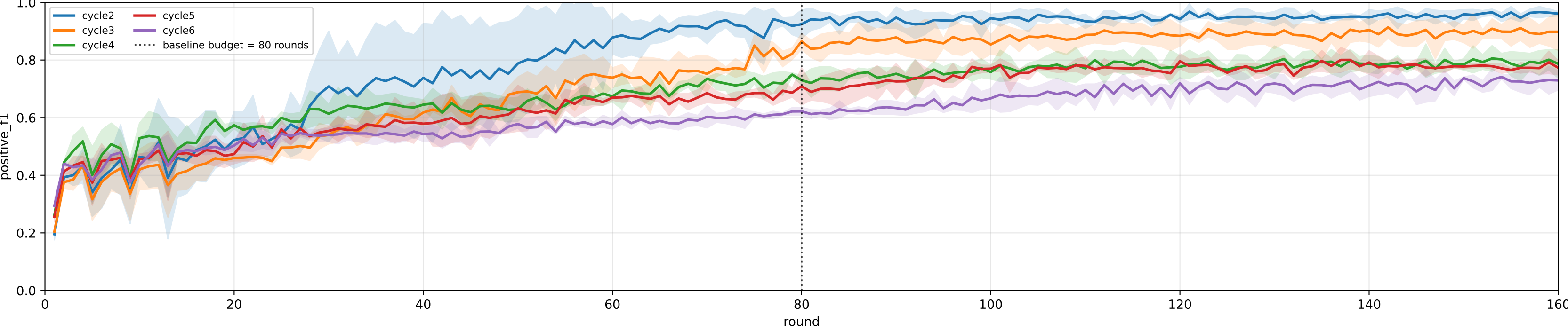
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



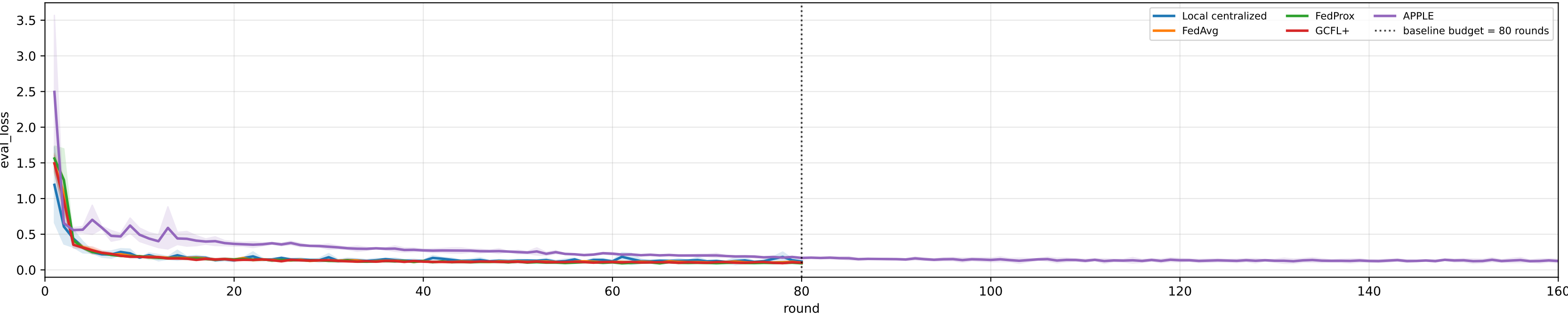
Controlled task-specialization level: Low | family aggregated / pooled performance |
family=specialized_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

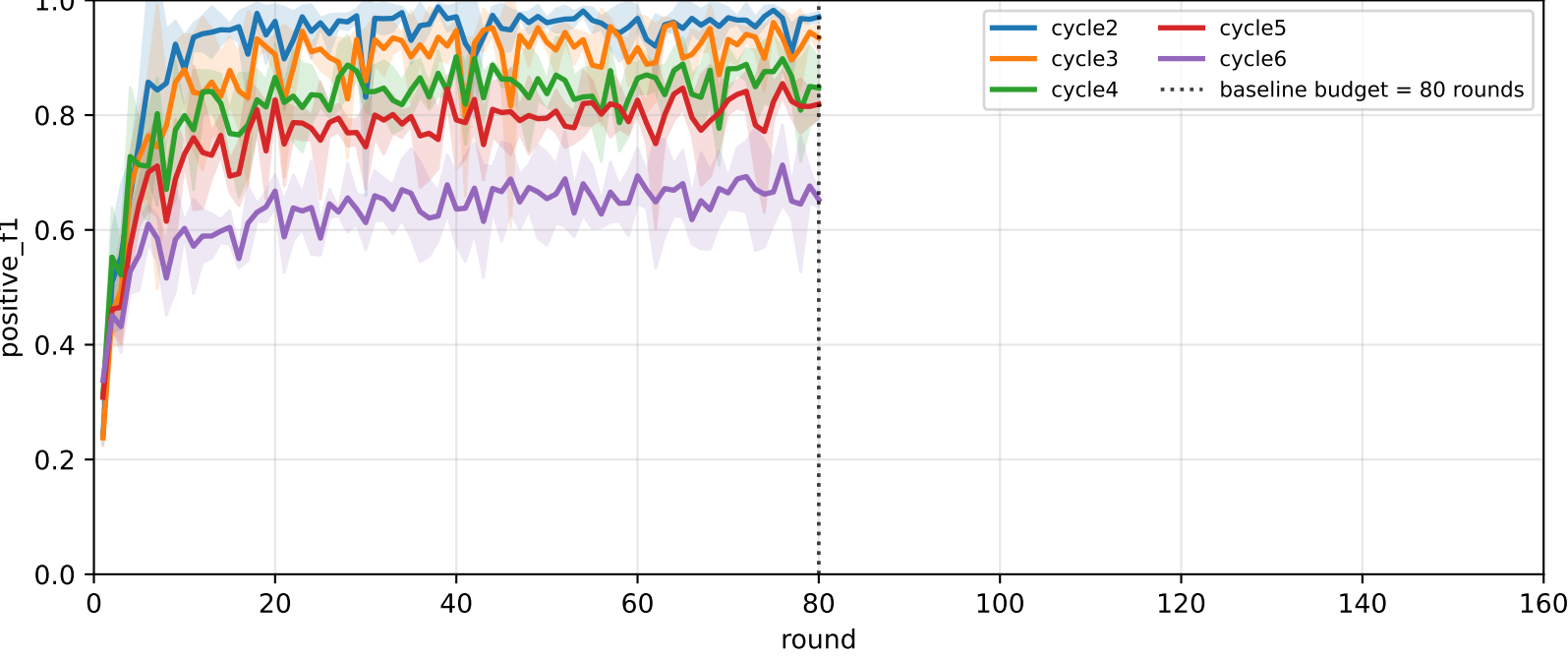
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.9% ± 2.9%	84.6% ± 2.1%	96.4% ± 1.2%	96.1% ± 0.4%	85.9% ± 3.7%	76.4% ± 0.2%	68.1% ± 7.4%
FedAvg	81.5% ± 0.6%	84.0% ± 0.3%	96.2% ± 1.0%	94.5% ± 0.8%	82.6% ± 2.1%	78.3% ± 0.6%	68.1% ± 3.3%
FedProx	83.7% ± 0.2%	85.5% ± 0.2%	96.0% ± 1.4%	94.6% ± 1.0%	85.7% ± 1.2%	78.2% ± 1.3%	73.1% ± 0.7%
GCFL+	82.0% ± 0.2%	83.9% ± 0.0%	95.2% ± 1.2%	93.3% ± 1.2%	83.3% ± 0.7%	77.0% ± 1.2%	70.8% ± 0.6%
APPLE	80.4% ± 4.0%	82.5% ± 3.6%	96.0% ± 1.4%	90.4% ± 3.6%	81.5% ± 5.9%	74.6% ± 2.9%	70.0% ± 5.6%

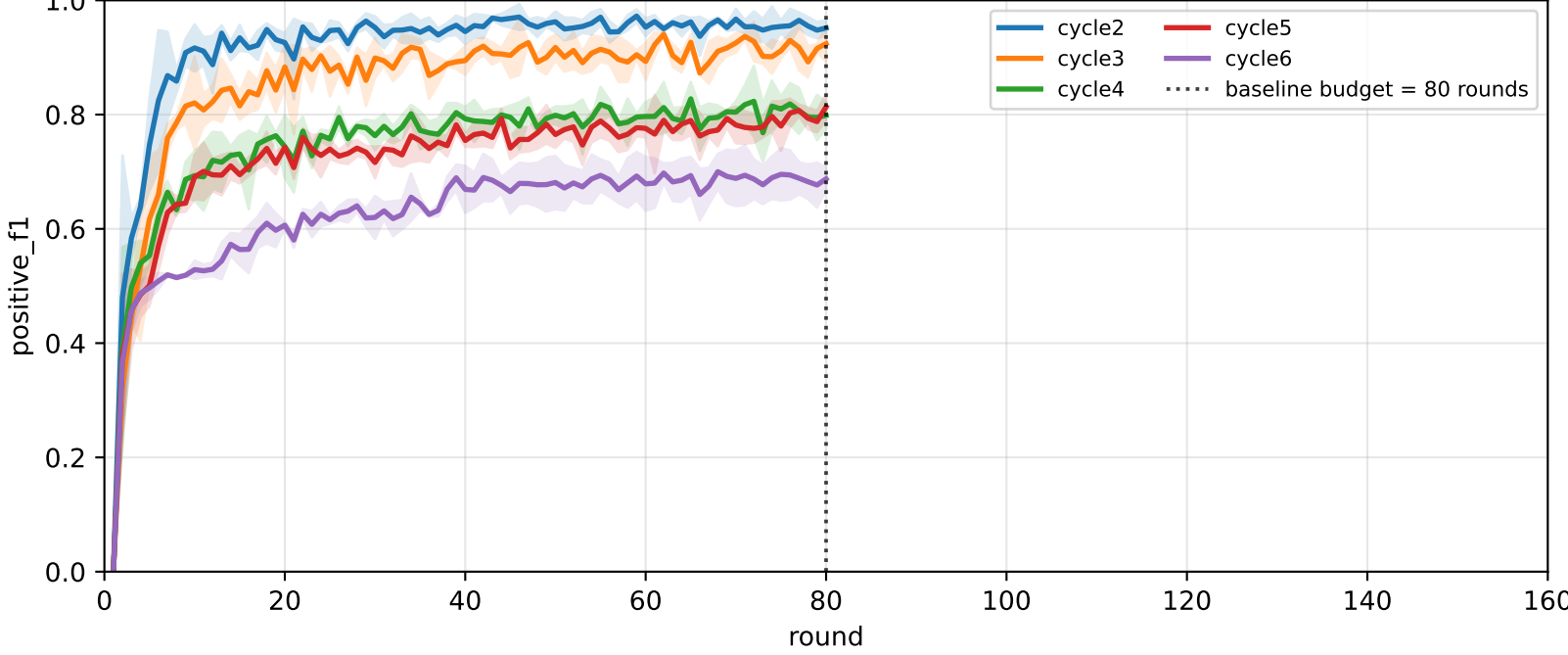
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



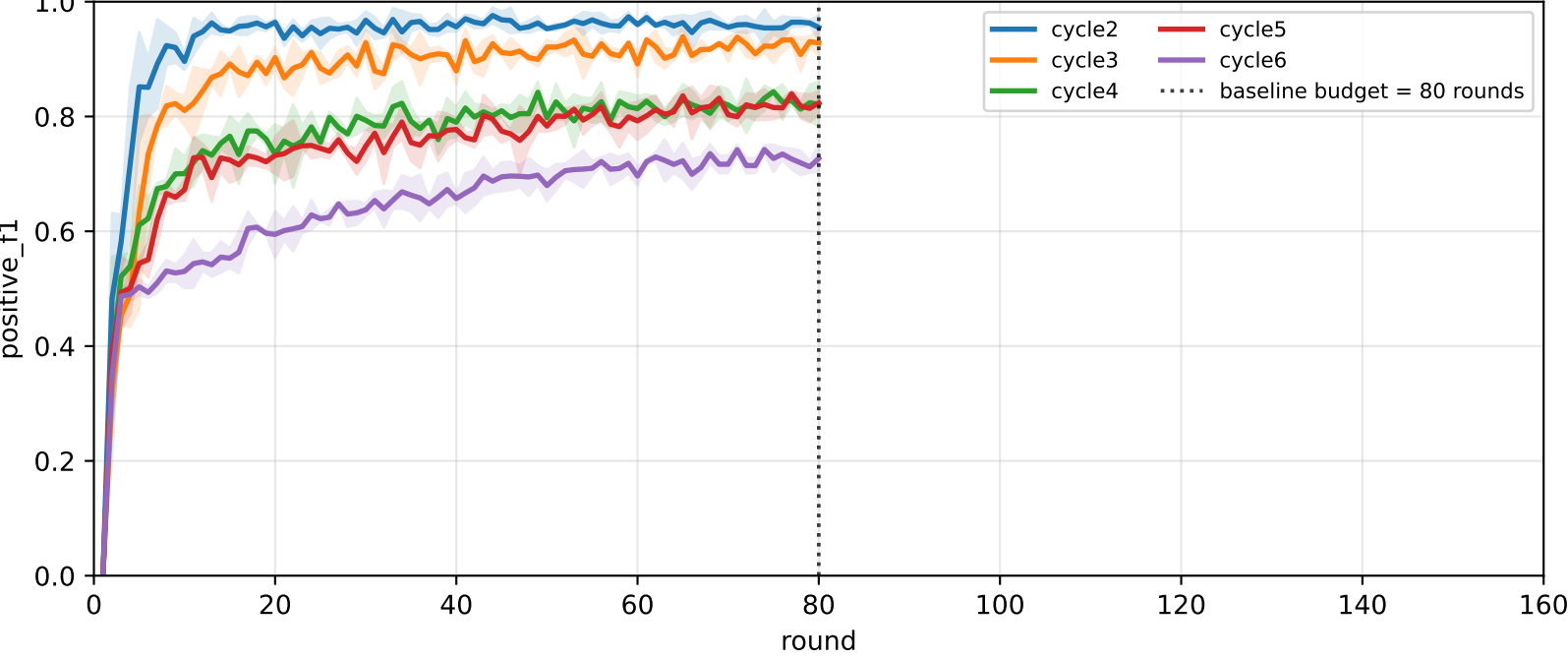
Pooled local centralized positive F1



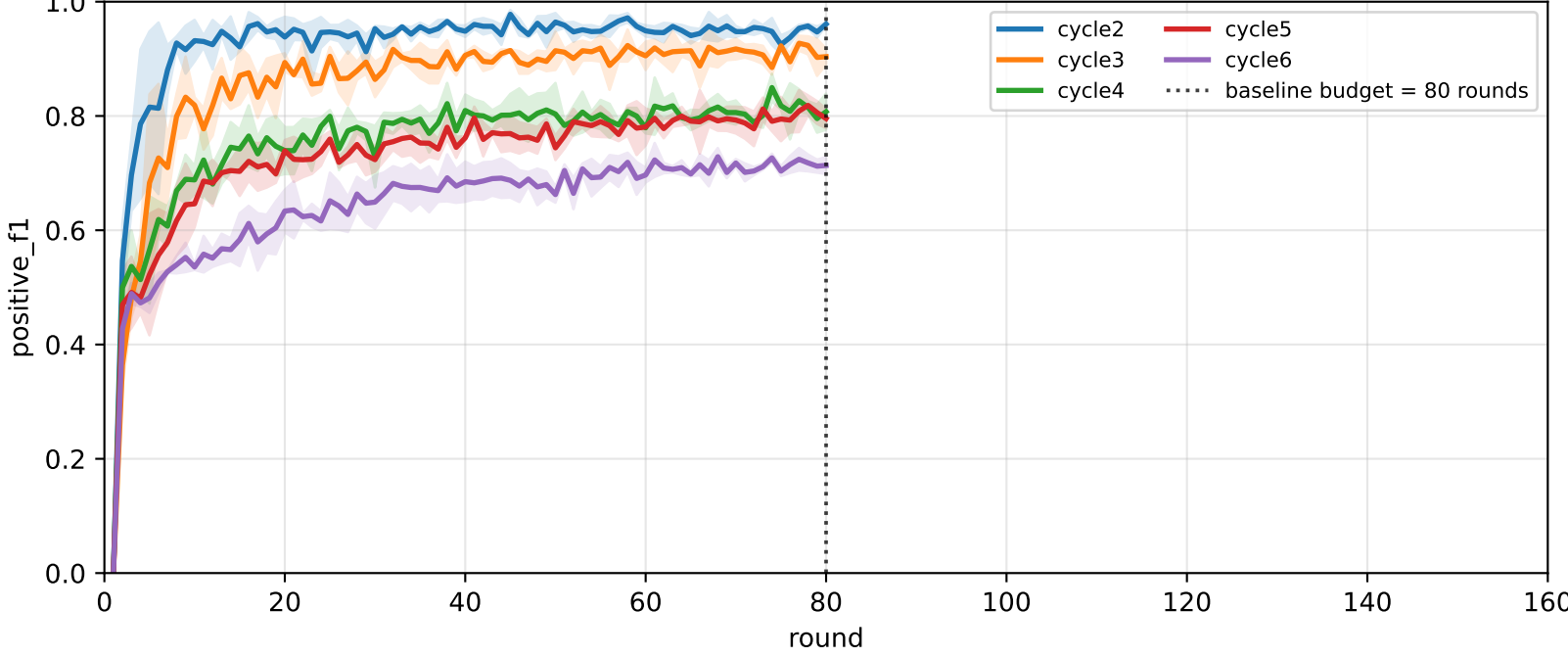
Pooled FedAvg positive F1



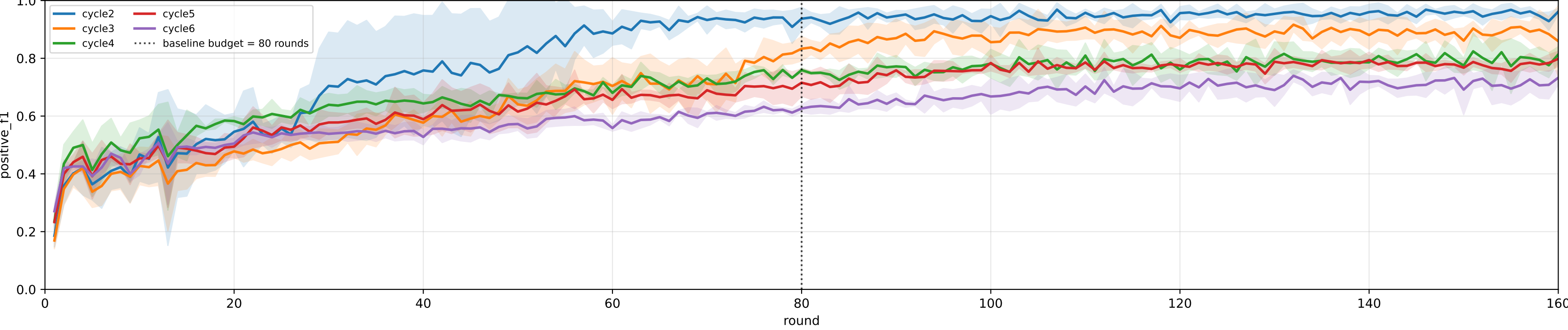
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



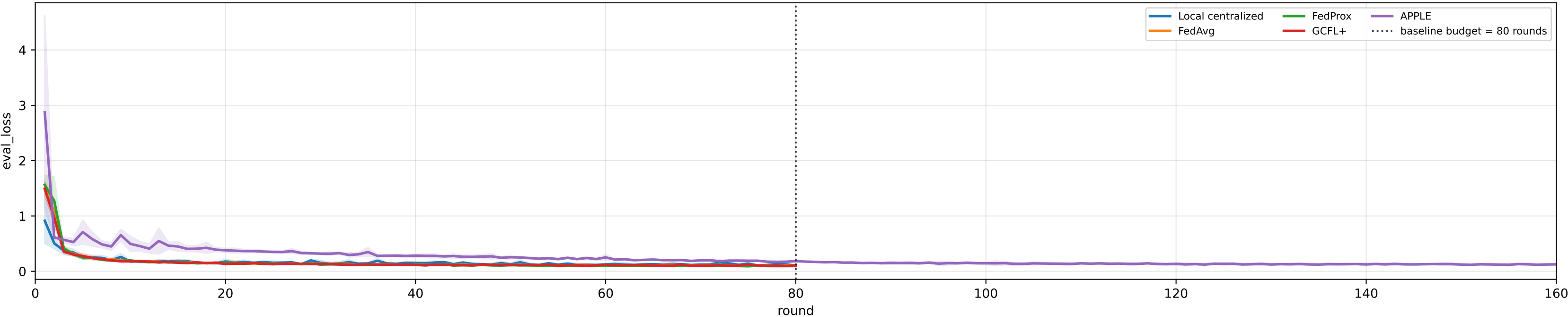
Controlled task-specialization level: Low | family aggregated / pooled performance |
family=specialized_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

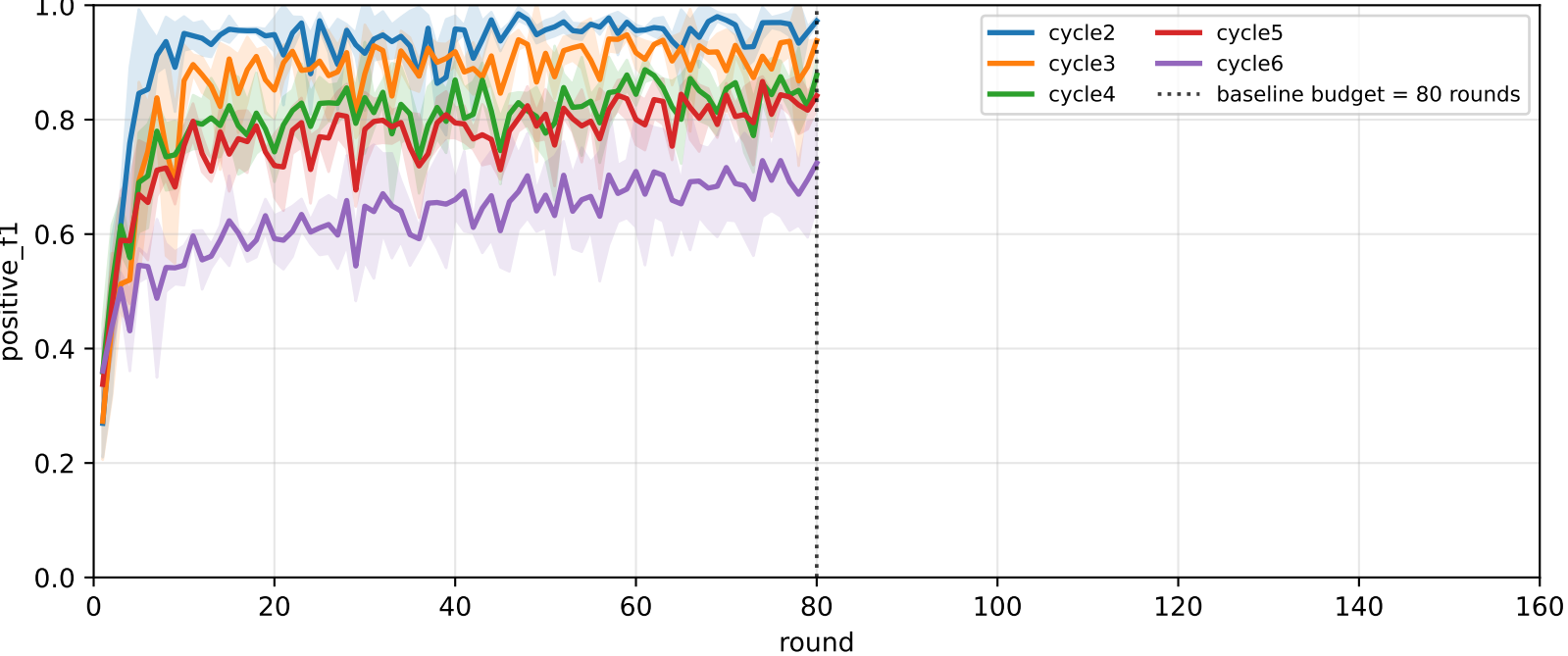
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	85.4% ± 1.5%	87.2% ± 0.5%	96.5% ± 1.7%	95.7% ± 0.7%	88.3% ± 2.6%	81.1% ± 0.7%	74.6% ± 7.4%
FedAvg	81.2% ± 0.3%	83.5% ± 0.4%	95.1% ± 1.1%	94.9% ± 1.1%	81.6% ± 3.1%	77.8% ± 0.9%	68.3% ± 2.6%
FedProx	82.9% ± 0.5%	84.6% ± 0.5%	93.9% ± 0.6%	94.1% ± 1.4%	84.5% ± 1.2%	77.7% ± 0.9%	72.5% ± 1.1%
GCFL+	81.7% ± 0.5%	83.8% ± 0.5%	95.0% ± 1.7%	93.6% ± 1.2%	83.1% ± 0.7%	77.2% ± 1.3%	69.9% ± 1.0%
APPLE	80.2% ± 1.3%	82.1% ± 1.1%	95.2% ± 1.8%	91.0% ± 1.6%	79.2% ± 1.8%	74.9% ± 1.7%	70.4% ± 1.9%

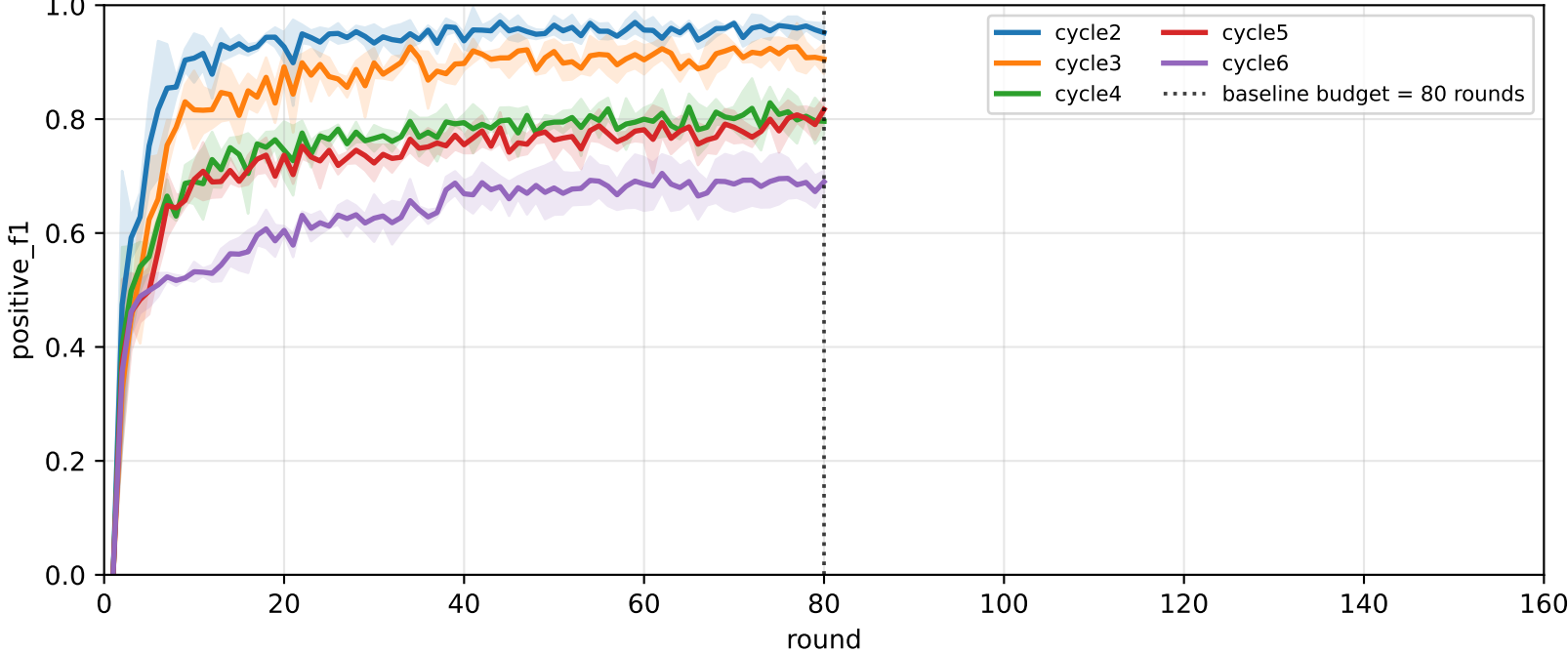
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



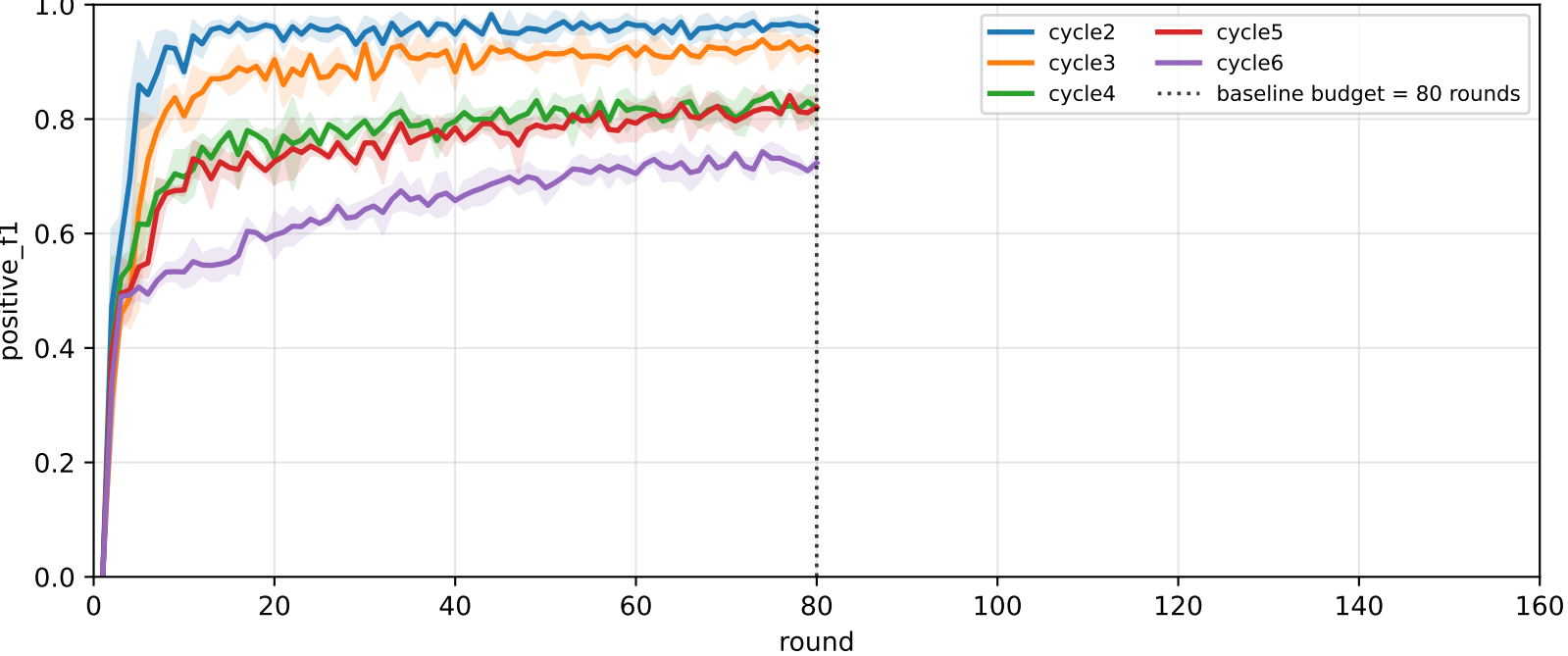
Pooled local centralized positive F1



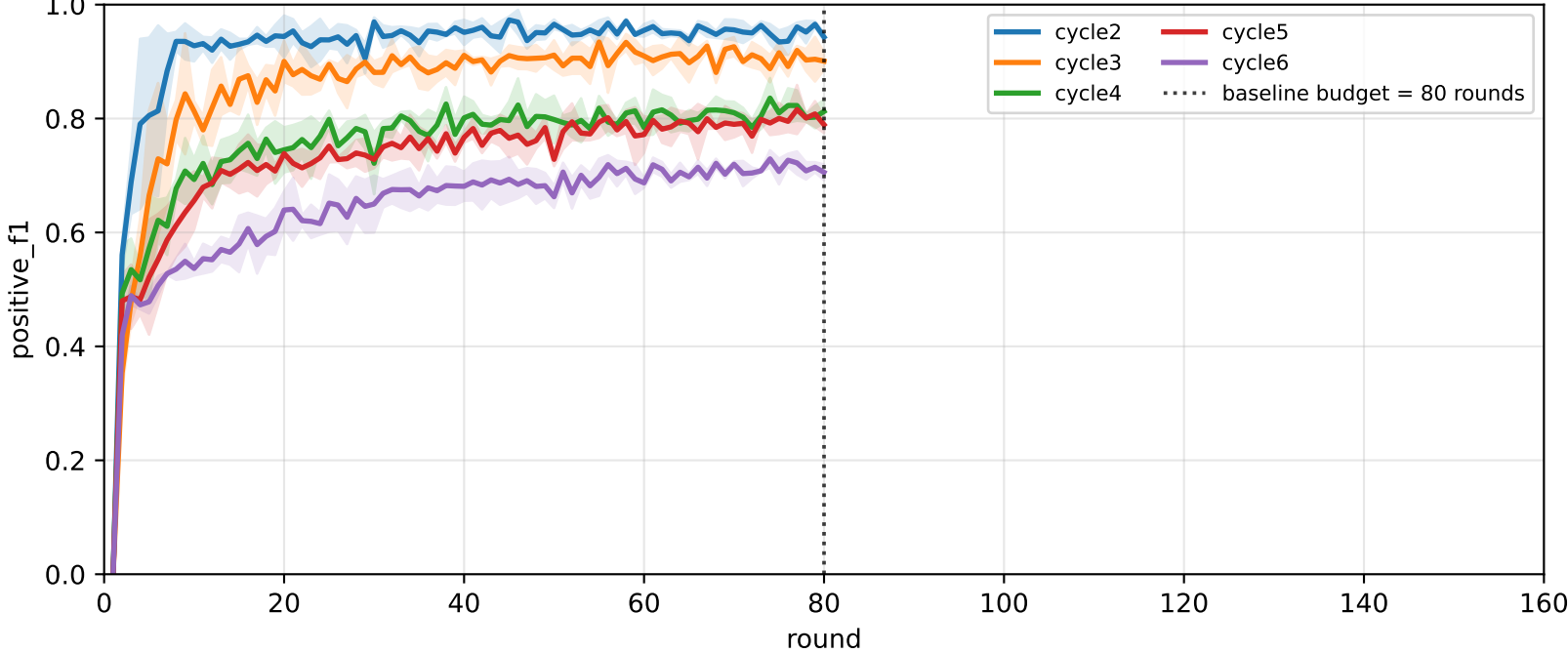
Pooled FedAvg positive F1



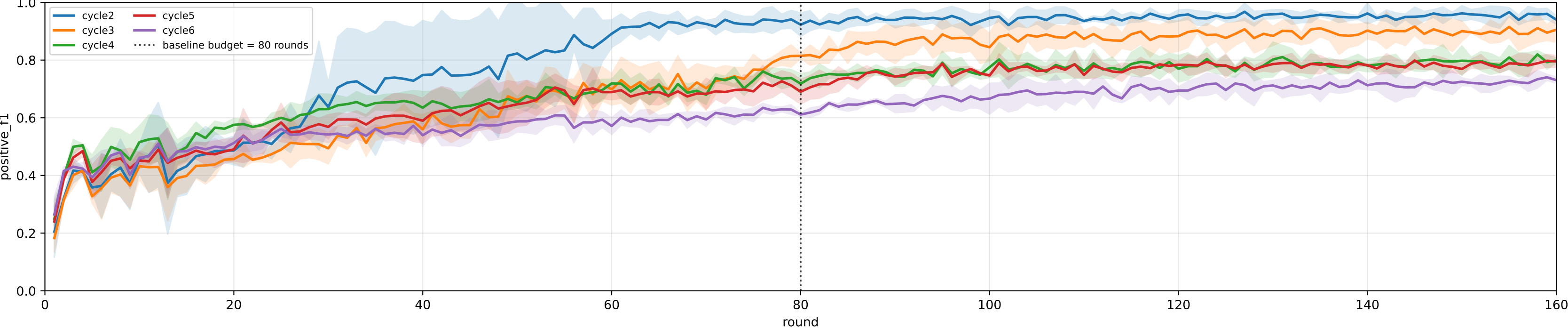
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



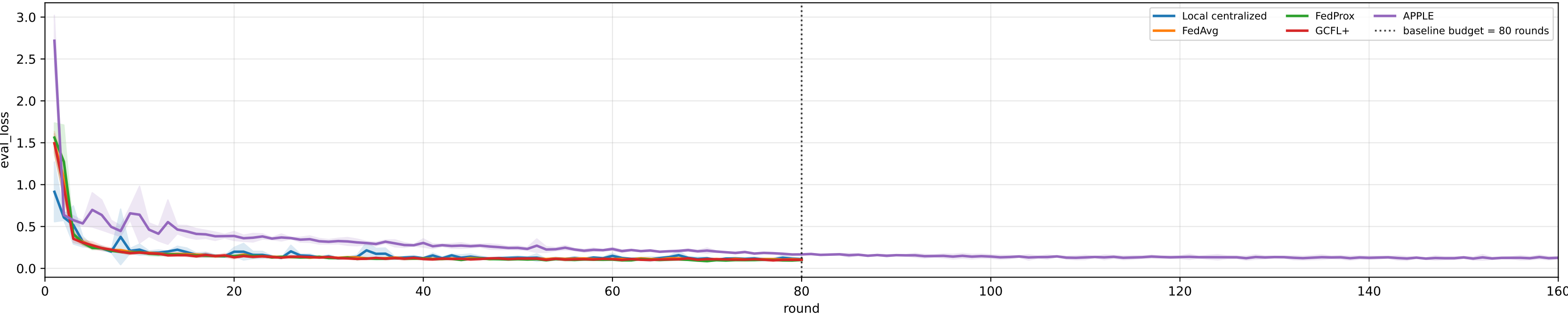
Controlled task-specialization level: Low | family aggregated / pooled performance |
family=specialized_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

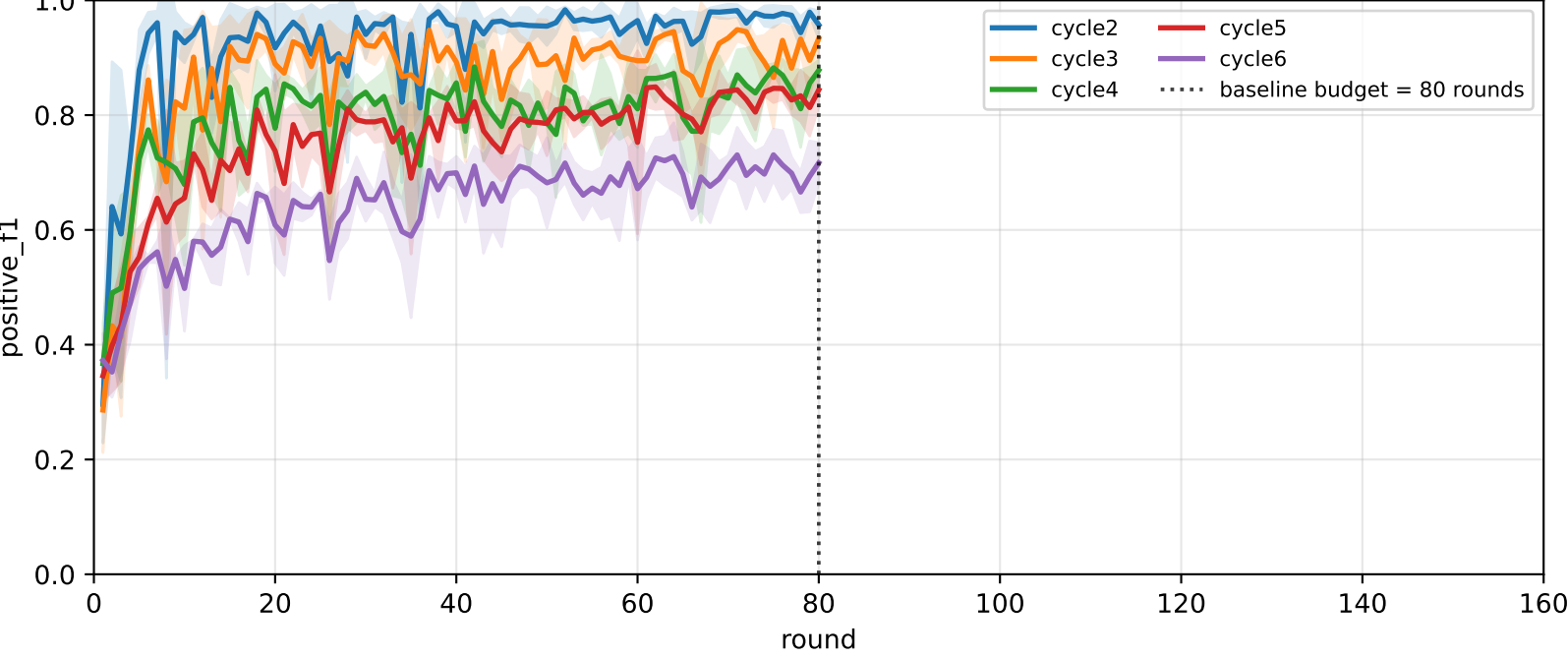
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.2% ± 1.5%	86.2% ± 1.0%	97.7% ± 0.7%	93.8% ± 0.8%	88.3% ± 1.2%	79.5% ± 0.8%	71.9% ± 4.5%
FedAvg	82.2% ± 0.7%	84.7% ± 0.3%	97.4% ± 0.3%	96.0% ± 1.1%	83.2% ± 2.3%	78.5% ± 0.9%	68.6% ± 2.8%
FedProx	83.8% ± 0.3%	85.7% ± 0.1%	95.5% ± 1.4%	95.8% ± 1.1%	85.9% ± 1.1%	78.2% ± 0.7%	72.9% ± 1.6%
GCFL+	81.7% ± 0.2%	83.8% ± 0.3%	95.4% ± 1.1%	93.6% ± 0.8%	82.9% ± 0.9%	76.9% ± 1.2%	69.8% ± 0.5%
APPLE	80.1% ± 1.3%	82.0% ± 1.3%	94.6% ± 1.8%	90.0% ± 1.8%	80.3% ± 2.0%	74.2% ± 0.7%	70.8% ± 1.8%

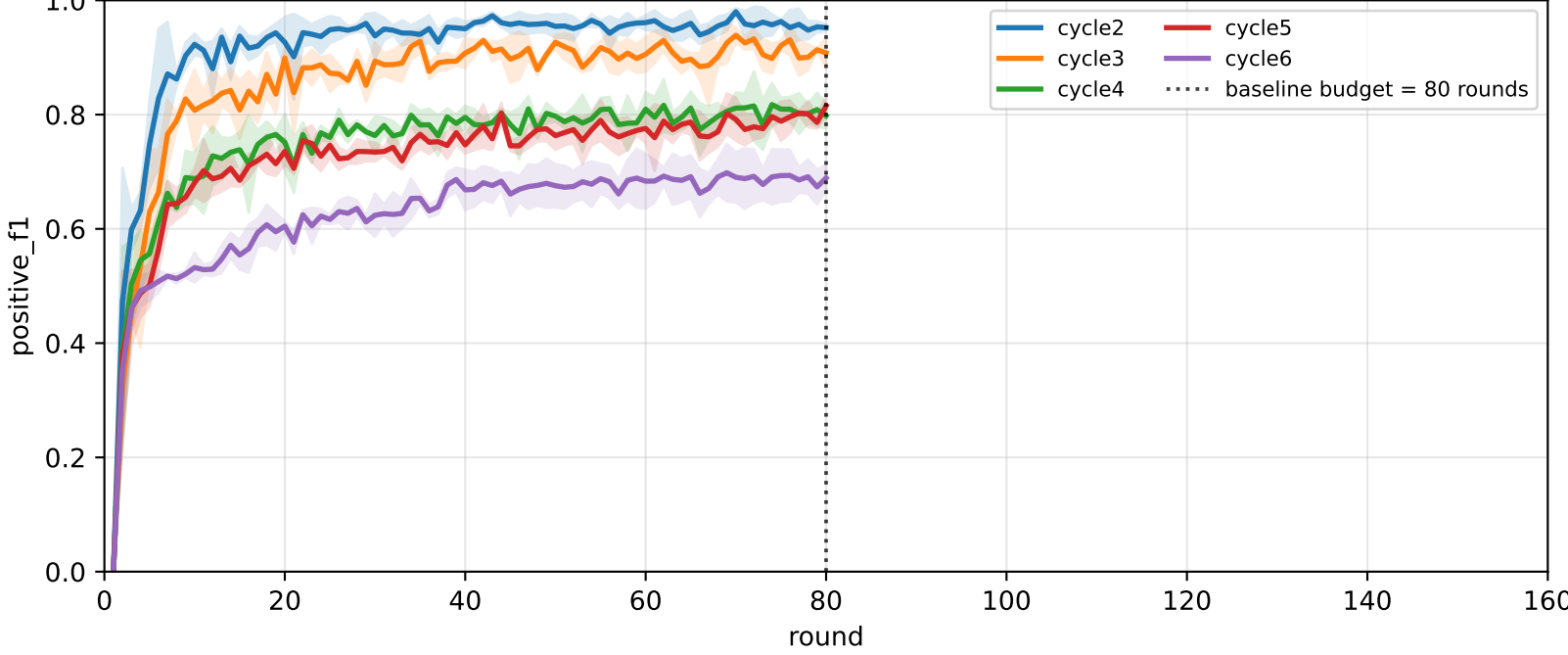
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



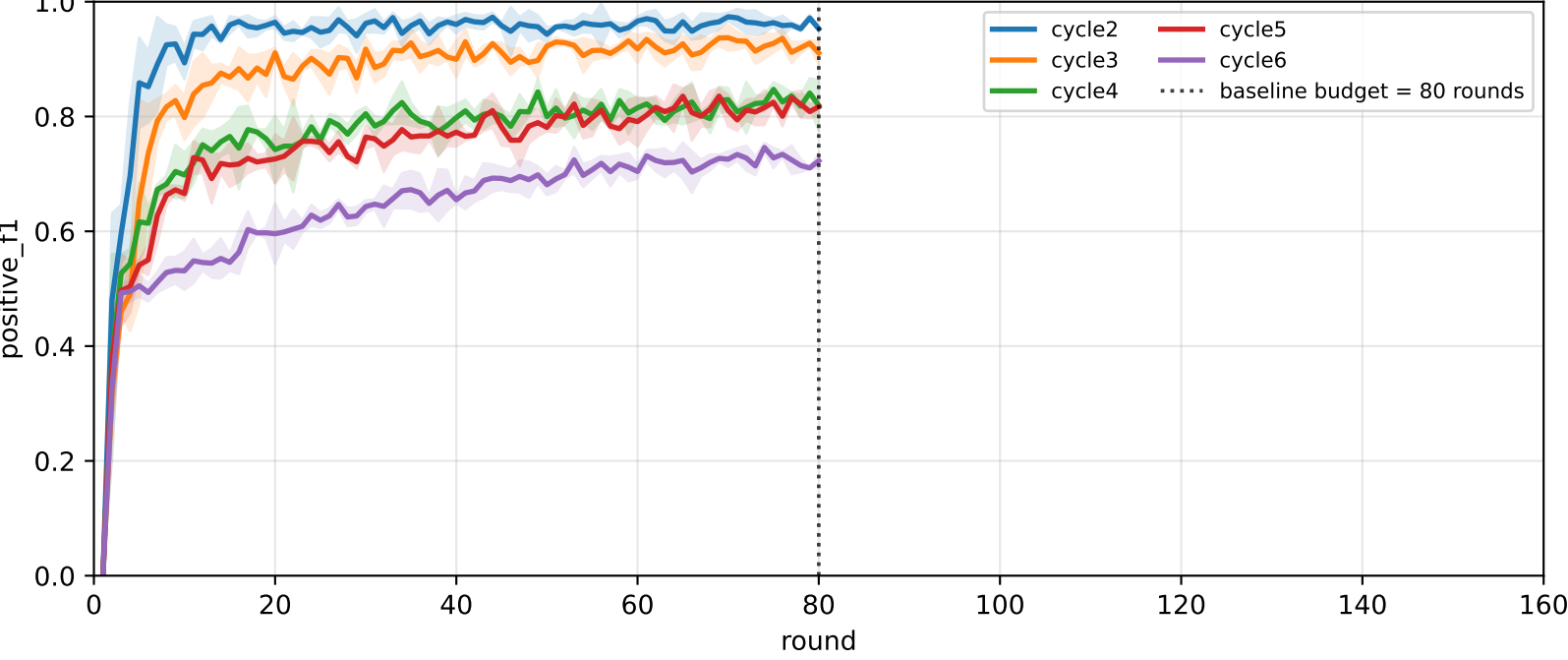
Pooled local centralized positive F1



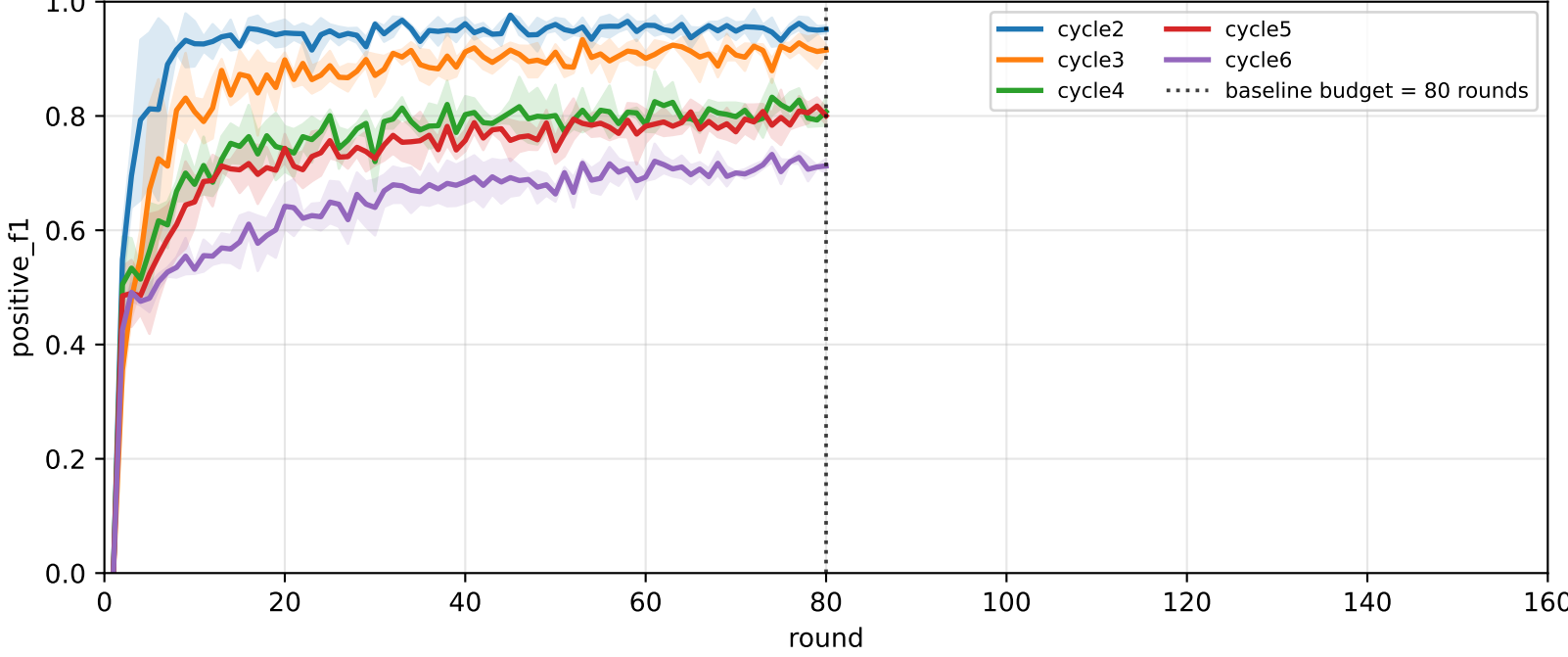
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)

